1.14 Mean Value

Worked Solutions

1 a)

$$\frac{1}{8-3}\int_3^8 (5x-2)\,\mathrm{d}x$$
$$= \frac{1}{5}\left[\frac{5}{2}x^2 - 2x\right]_3^8 = \frac{1}{5}\left[(160-16) - \left(\frac{45}{2} - 6\right)\right] = \frac{51}{2}$$

1 b)

$$\frac{1}{5-2}\int_2^5 5x^2\,\mathrm{d}x$$
$$= \frac{1}{3}\left[\frac{5}{3}x^3\right]_2^5 = \frac{1}{3}\left[\frac{625}{3} - \frac{40}{3}\right] = 65$$

1 c)

$$\frac{1}{16-9}\int_9^{16} \frac{1}{\sqrt{x}}\,\mathrm{d}x = \frac{1}{7}\int_9^{16} x^{-\frac{1}{2}}\,\mathrm{d}x$$
$$= \frac{1}{7}\left[2x^{\frac{1}{2}}\right]_9^{16} = \frac{1}{7}[8-6] = \frac{2}{7}$$

2)

$$\frac{1}{1.2 - -0.5}\int_{-0.5}^{1.2} (2x^2-1)\,\mathrm{d}x$$
$$= \frac{10}{17}\left[\frac{2}{3}x^3 - x\right]_{-0.5}^{1.2}$$
$$= \frac{10}{17}\left[\left(\frac{144}{125} - \frac{6}{5}\right) - \left(-\frac{1}{12} + \frac{1}{2}\right)\right] = -\frac{41}{150} = -0.273 \text{ (3 SF)}$$

3)

$$\frac{1}{k-0}\int_0^k x^2\,\mathrm{d}x$$
$$= \frac{1}{k}\left[\frac{1}{3}x^3\right]_0^k = \frac{1}{k}\left[\left(\frac{1}{3}k^3\right) - (0)\right]$$
$$= \frac{1}{3}k^2 \text{ as req.}$$

4)

$$\frac{1}{a-0}\int_0^a 8x^3\,\mathrm{d}x$$
$$= \frac{1}{a}[2x^4]_0^a = \frac{1}{a}[2a^4 - 0] = 2a^3$$

5)

$$\frac{1}{b-a}\int_a^b (mx+c)\,\mathrm{d}x$$
$$= \frac{1}{b-a}\left[\frac{1}{2}mx^2 + cx\right]_a^b = \frac{1}{b-a}\left[\left(\frac{1}{2}mb^2 + bc\right) - \left(\frac{1}{2}ma^2 + ca\right)\right]$$
$$= \frac{1}{b-a}\left[\frac{1}{2}m(b^2-a^2) + c(b-a)\right]$$
$$= \frac{1}{b-a}\left[\frac{1}{2}m(b-a)(b+a) + c(b-a)\right]$$
$$= \frac{1}{2}m(b+a) + c \text{ as req.}$$

6)

$$\frac{1}{2a-a}\int_a^{2a}(x+4)^2\,\mathrm{d}x = \frac{1}{a}\int_a^{2a}(x^2+8x+16)\,\mathrm{d}x$$
$$= \frac{1}{a}\left[\frac{1}{3}x^3 + 4x^2 + 16x\right]_a^{2a}$$
$$= \frac{1}{a}\left[\left(\frac{1}{3}(2a)^3 + 4(2a)^2 + 16(2a)\right) - \left(\frac{1}{3}a^3 + 4a^2 + 16a\right)\right]$$
$$= \frac{1}{a}\left(\frac{8}{3}a^3 + 16a^2 + 32a - \frac{1}{3}a^3 - 4a^2 - 16a\right)$$
$$= \frac{7}{3}a^2 + 12a + 16$$

7)

$$\frac{1}{k-0}\int_0^k \frac{1+x}{\sqrt{x}}\,\mathrm{d}x = \frac{1}{k}\int_0^k \left(x^{-\frac{1}{2}} + x^{\frac{1}{2}}\right)\,\mathrm{d}x$$
$$= \frac{1}{k}\left[2x^{\frac{1}{2}} + \frac{2}{3}x^{\frac{3}{2}}\right]_0^k$$
$$= \frac{1}{k}\left[\left(2k^{\frac{1}{2}} + \frac{2}{3}k^{\frac{3}{2}}\right) - (0)\right]$$
$$= 2k^{-\frac{1}{2}} + \frac{2}{3}k^{\frac{1}{2}}$$
$$= \frac{2}{\sqrt{k}} + \frac{2\sqrt{k}}{3}$$

8)

$$\begin{aligned}
&\frac{1}{9-1}\int_1^9 (2\sqrt{x}+k)\,\mathrm{d}x = \frac{1}{8}\int_1^9 \left(2x^{\frac{1}{2}}+k\right)\,\mathrm{d}x \\
&= \frac{1}{8}\left[\frac{4}{3}x^{\frac{3}{2}}+kx\right]_1^9 \\
&= \frac{1}{8}\left[(36+9k)-\left(\frac{4}{3}+k\right)\right] \\
&= k+\frac{13}{3}
\end{aligned}$$

Since $k+\frac{13}{3}=\frac{22}{3}$ then $k=3$

9)

$$\begin{aligned}
&\frac{1}{5-2}\int_2^5 (x^2+1)\,\mathrm{d}x \\
&= \frac{1}{3}\left[\frac{1}{3}x^3+x\right]_2^5 \\
&= \frac{1}{3}\left[\left(\frac{125}{3}+5\right)-\left(\frac{8}{3}+2\right)\right] \\
&= 14
\end{aligned}$$

So that $\frac{1}{k-0}\int_0^k (x^2+1)\,\mathrm{d}x = 28$

$$\begin{aligned}
\frac{1}{k}\left[\frac{1}{3}x^3+x\right]_0^k &= 28 \\
\frac{1}{k}\left[\left(\frac{k^3}{3}+k\right)-0\right] &= 28 \\
\frac{k^2}{3}+1 &= 28
\end{aligned}$$

So $k=\pm 9$ $\therefore$ since k is positive, $k=9$

10)

$$\begin{aligned}
&\frac{1}{k-0}\int_0^k (3x^2+12x-40)\,\mathrm{d}x \\
&= \frac{1}{k}[x^3+6x^2-40x]_0^k \\
&= \frac{1}{k}[(k^3+6k^2-40k)-0] \\
&= k^2+6k-40
\end{aligned}$$

Since $k^2+6k-40=51$ then

$$\begin{aligned}
k^2+6k-91 &= 0 \\
(k-7)(k+13) &= 0
\end{aligned}$$

Since $k>0$ then $k=7$

11 a)

$$\begin{aligned}
&\frac{1}{1+2}\int_{-2}^{1}(ax-4)\,\mathrm{d}x \\
&=\frac{1}{3}\left[\frac{1}{2}ax^2-4x\right]_{-2}^{1} \\
&=\frac{1}{3}\left[\left(\frac{1}{2}a-4\right)-(2a+8)\right] \\
&=\frac{1}{3}\left[-\frac{3}{2}a-12\right] \\
&=-\frac{1}{2}a-4
\end{aligned}$$

And so $-\frac{1}{2}a-4=-5.5\Rightarrow a=3$

11 b)

$$\begin{aligned}
&\frac{1}{2p-p}\int_{p}^{2p}(3x-4)\,\mathrm{d}x \\
&=\frac{1}{p}\left[\frac{3}{2}x^2-4x\right]_{p}^{2p} \\
&=\frac{1}{p}\left[\left(\frac{3}{2}(2p)^2-4(2p)\right)-\left(\frac{3}{2}p^2-4p\right)\right] \\
&=\frac{1}{p}\left[\frac{9}{2}p^2-4p\right] \\
&=\frac{9}{2}p-4
\end{aligned}$$

And $\frac{9}{2}p-4=18.5\Rightarrow p=5$

12 a)

$$\frac{1}{3-1}\int_1^3\frac{1}{6}\sqrt{t}\left(12-\frac{1}{t}\right)\,\mathrm{d}t=\frac{1}{2}\int_1^3\left(2t^{\frac{1}{2}}-\frac{1}{6}t^{-\frac{1}{2}}\right)\,\mathrm{d}t$$

$$=\frac{1}{2}\left[\frac{4}{3}t^{\frac{3}{2}}-\frac{1}{3}t^{\frac{1}{2}}\right]_1^3$$

$$=\frac{1}{2}\left[\left(\frac{4}{3}\times 3^{\frac{3}{2}}-\frac{1}{3}\times 3^{\frac{1}{2}}\right)-\left(\frac{4}{3}-\frac{1}{3}\right)\right]=2.675426\ldots=2.68\ (3\text{ SF})\text{ as req.}$$

12 b)

$$v=2t^{\frac{1}{2}}-\frac{1}{6}t^{-\frac{1}{2}}$$

$$a=\frac{\mathrm{d}v}{\mathrm{d}t}=t^{-\frac{1}{2}}+\frac{1}{12}t^{-\frac{3}{2}}$$

$$\frac{1}{3-1}\int_1^3\left(t^{-\frac{1}{2}}+\frac{1}{12}t^{-\frac{3}{2}}\right)\,\mathrm{d}t$$

$$= \frac{1}{2}\left[2x^{\frac{1}{2}} - \frac{1}{6}x^{-\frac{1}{2}}\right]_1^3 = \frac{1}{2}\left[\left(2\times 3^{\frac{1}{2}} - \frac{1}{6}\times 3^{-\frac{1}{2}}\right) - \left(2 - \frac{1}{6}\right)\right] = 0.7672716\ldots$$
$$= 0.767 \text{ ms}^{-2} \text{ (3 SF)}$$

13)

$$\frac{1}{3-0}\int_0^3 (-5x^3 + 12x^2 + 4x)\,\mathrm{d}x$$
$$= \frac{1}{3}\left[-\frac{5}{4}x^4 + 4x^3 + 2x^2\right]_0^3 = \frac{1}{3}\left[\left(-\frac{405}{4} + 108 + 18\right) - 0\right] = 8.25\ ^\circ\text{C}$$

14)

$$\frac{1}{12-0}\int_0^{12} (2t^3 - 7t^2 + 15t + 40)\,\mathrm{d}t$$
$$= \frac{1}{12}\left[\frac{1}{2}t^4 - \frac{7}{3}t^3 + \frac{15}{2}t^2 + 40t\right]_0^{12} = \frac{1}{12}[(10368 - 4032 + 1080 + 480) - 0]$$
$$= 658 \text{ rabbits}$$

15)

$$\frac{1}{5-2}\int_2^5 (x^3 - 6x^2 + 9x + 4)\,\mathrm{d}x$$
$$= \frac{1}{3}\left[\frac{1}{4}x^4 - 2x^3 + \frac{9}{2}x^2 + 4x\right]_2^5$$
$$= \frac{1}{3}\left[\left(\frac{625}{4} - 250 + \frac{225}{2} + 20\right) - (4 - 16 + 18 + 8)\right] = \frac{33}{4}$$

$\therefore k = \frac{33}{4}$

16)

$$\frac{1}{3-1}\int_1^3 (x^3 - 3x^2 - 9x + 10)\,\mathrm{d}x$$
$$= \frac{1}{2}\left[\frac{1}{4}x^4 - x^3 - \frac{9}{2}x^2 + 10x\right]_1^3$$
$$= \frac{1}{2}\left[\left(\frac{81}{4} - 27 - \frac{81}{2} + 30\right) - \left(\frac{1}{4} - 1 - \frac{9}{2} + 10\right)\right] = -11$$

$\therefore k = -11$

17)

$$\frac{1}{k-2}\int_2^k (x^3 - 2)\,\mathrm{d}x$$

$$
\begin{aligned}
&= \frac{1}{k-2}\left[\frac{1}{4}x^4 - 2x\right]_2^k \\
&= \frac{1}{k-2}\left[\left(\frac{1}{4}k^4 - 2k\right) - (4-4)\right] \\
&= \frac{1}{k-2}\left(\frac{1}{4}k^4 - 2k\right)
\end{aligned}
$$

Since $\frac{1}{k-2}\left(\frac{1}{4}k^4 - 2k\right) = \frac{195}{4}$ then

$$
\begin{aligned}
4\left(\frac{1}{4}k^4 - 2k\right) &= 195(k-2) \\
k^4 - 8k &= 195k - 390 \\
k^4 - 203k + 390 &= 0
\end{aligned}
$$

Let $\mathrm{p}(k) = k^4 - 203k + 390$

$$
\begin{aligned}
\mathrm{p}(5) &= 625 - 1015 + 390 = 0 \\
\mathrm{p}(2) &= 16 - 406 + 390 = 0
\end{aligned}
$$

So $k = 2$ and $k = 5$ is a solution.

$$k^4 - 203k + 390 = (k-5)(k-2)(k^2 + 7k + 39)$$

k needs to be real and > 2

$\therefore k = 5$

18 a)

$$
\begin{aligned}
&\frac{1}{4-2}\int_2^4 (x^2 + x - 1)\,\mathrm{d}x \\
&= \frac{1}{2}\left[\frac{1}{3}x^3 + \frac{1}{2}x^2 - x\right]_2^4 = \frac{1}{2}\left[\left(\frac{64}{3} + 8 - 4\right) - \left(\frac{8}{3} + 2 - 2\right)\right] = \frac{34}{3}
\end{aligned}
$$

18 b)

$$25 - \frac{34}{3} = \frac{41}{3} \quad \therefore c = \frac{41}{3}$$

Alternative Method:

$$
\begin{aligned}
\frac{1}{4-2}\int_2^4 (x^2 + x - 1 + c)\,\mathrm{d}x &= 25 \\
\frac{1}{2}\left[\frac{1}{3}x^3 + \frac{1}{2}x^2 - x + cx\right]_2^4 &= 25 \\
\left(\frac{76}{3} + 4c\right) - \left(\frac{8}{3} + 2c\right) &= 50 \\
\frac{68}{3} + 2c &= 50 \\
\therefore c &= \frac{41}{3}
\end{aligned}
$$

19 a)

A horizontal translation does not affect the mean value, but a vertical one does, so the mean value of g is given by $k - 1$

Alternative Method:

$$\frac{1}{4}\int_0^4 \mathrm{f}(x)\,\mathrm{d}x = k$$

Since $\mathrm{g}(x) = \mathrm{f}(x-2) - 1$ the mean value of g is given by

$$\begin{aligned}
&\frac{1}{4}\int_{-2}^{2} \mathrm{g}(x)\,\mathrm{d}x = \frac{1}{4}\int_{-2}^{2} (\mathrm{f}(x-2) - 1)\,\mathrm{d}x \\
&= \frac{1}{4}\int_{-2}^{2} \mathrm{f}(x-2)\,\mathrm{d}x + \frac{1}{4}\int_{-2}^{2} -1\,\mathrm{d}x \\
&= \frac{1}{4}\int_{0}^{4} \mathrm{f}(x)\,\mathrm{d}x + \frac{1}{4}\int_{-2}^{2} -1\,\mathrm{d}x \\
&= k - 1
\end{aligned}$$

19 b)

There has been a reflection of all the y-values so the mean value of g is given by $-k$

Alternative Method:

$$\frac{1}{4}\int_0^4 \mathrm{f}(x)\,\mathrm{d}x = k$$

Since $\mathrm{g}(x) = -\mathrm{f}(x)$ the mean value of g is given by

$$\begin{aligned}
&\frac{1}{4}\int_{2}^{6} \mathrm{g}(x)\,\mathrm{d}x = \frac{1}{4}\int_{2}^{6} (-\mathrm{f}(x))\,\mathrm{d}x \\
&= -\frac{1}{4}\int_{2}^{6} \mathrm{f}(x)\,\mathrm{d}x \\
&= -k
\end{aligned}$$

19 c)

Due to symmetry in the intervals, the mean value has not changed, so the mean value of g is given by $k + 4$

Alternative Method:

$$\frac{1}{4}\int_0^4 \mathrm{f}(x)\,\mathrm{d}x = k$$

Since $\mathrm{m}(x) = \mathrm{f}(-x)$ the mean value of g is given by

$$\begin{aligned}
&\frac{1}{4}\int_{-4}^{0} \mathrm{m}(x)\,\mathrm{d}x = \frac{1}{4}\int_{-4}^{0} \mathrm{f}(-x)\,\mathrm{d}x \\
&= \frac{1}{4}\int_{0}^{4} \mathrm{f}(x)\,\mathrm{d}x \\
&= k
\end{aligned}$$

19 d)

The horizontal translation and symmetry mean the mean value does not change, but the vertical translation does affect the mean value. The mean value of g is given by $k + 3$

Alternative Method:

$$\frac{1}{4}\int_0^4 \mathrm{f}(x)\,\mathrm{d}x = k$$

Since $\mathrm{n}(x) = \mathrm{m}(x+1) + 3$ and $\mathrm{m}(x) = \mathrm{f}(-x)$, the mean value of n is given by

$$\begin{aligned}
&\frac{1}{4}\int_{-3}^1 \mathrm{n}(x)\,\mathrm{d}x = \frac{1}{4}\int_{-3}^1 (\mathrm{m}(x+1)+3)\,\mathrm{d}x \\
&= \frac{1}{4}\int_{-3}^1 \mathrm{m}(x+1)\,\mathrm{d}x + \frac{1}{4}\int_{-3}^1 3\,\mathrm{d}x \\
&= \frac{1}{4}\int_0^4 \mathrm{f}(x)\,\mathrm{d}x + \frac{1}{4}\int_{-3}^1 3\,\mathrm{d}x \\
&= k+3
\end{aligned}$$

20 a)

A horizontal translation does not affect the mean value, but a vertical one does, so the mean value of g is given by $k+4$

Alternative Method:

$$\frac{1}{5}\int_2^7 \mathrm{f}(x)\,\mathrm{d}x = k$$

Since $\mathrm{g}(x) = \mathrm{f}(x+1) + 4$ the mean value of g is given by

$$\begin{aligned}
&\frac{1}{5}\int_1^6 \mathrm{g}(x)\,\mathrm{d}x = \frac{1}{5}\int_1^6 (\mathrm{f}(x+1)+4)\,\mathrm{d}x \\
&= \frac{1}{5}\int_1^6 \mathrm{f}(x+1)\,\mathrm{d}x + \frac{1}{5}\int_1^6 4\,\mathrm{d}x \\
&= \frac{1}{5}\int_2^7 \mathrm{f}(x)\,\mathrm{d}x + \frac{1}{5}\int_1^6 4\,\mathrm{d}x \\
&= k+4
\end{aligned}$$

20 b)

There has been a stretch of all the x-values so the mean value of g is given by k

Alternative Method:

$$\frac{1}{5}\int_2^7 \mathrm{f}(x)\,\mathrm{d}x = k$$

Since $\mathrm{g}(x) = \mathrm{f}(3x)$ the mean value of g is given by

$$\begin{aligned}
&\frac{1}{5}\int_6^{21} \mathrm{f}(3x)\,\mathrm{d}x = \frac{1}{5}\int_2^7 \mathrm{f}(x)\,\mathrm{d}x \\
&= k
\end{aligned}$$

20 c)

There has been a stretch of all the y-values so the mean value of g is given by $\frac{1}{2}k$

Alternative Method:

$$\frac{1}{5}\int_2^7 \mathrm{f}(x)\,\mathrm{d}x = k$$

Since $\mathrm{m}(x) = \frac{1}{2}\mathrm{f}(x)$ the mean value of g is given by

$$\frac{1}{5}\int_2^7 \mathrm{m}(x)\,\mathrm{d}x = \frac{1}{4}\int_2^7 \frac{1}{2}\mathrm{f}(x)\,\mathrm{d}x$$

$$= \frac{1}{2} \times \frac{1}{5} \int_2^7 \mathrm{f}(x)\,\mathrm{d}x$$
$$= \frac{1}{2}k$$

21)

$$[\mathrm{f}(x)]^2 = [-x(x-0.5)(x-1)]^2 = x^2(x-0.5)^2(x-1)^2$$
$$= x^2(x^2 - x + 0.25)(x^2 - 2x + 1)$$
$$= x^2(x^4 - 3x^3 + 3.25x^2 - 1.5x + 0.25)$$
$$= x^6 - 3x^5 + 3.25x^4 - 1.5x^3 + 0.25x^2$$

$$\frac{1}{0.6 - 0.1} \int_{0.1}^{0.6} (x^6 - 3x^5 + 3.25x^4 - 1.5x^3 + 0.25x^2)\,\mathrm{d}x$$
$$= 2\left[\frac{1}{7}x^7 - \frac{1}{2}x^6 + \frac{13}{20}x^5 - \frac{3}{8}x^4 + \frac{1}{12}x^3\right]_{0.1}^{0.6}$$
$$= 0.00112648$$

$$\sqrt{0.00112648} = 0.0336 \text{ (3 SF)}$$